

Guía 3 - Ej 8

Ecuaciones de equilibrio diferenciales

$$1) \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + X = 0, 2) \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + Y = 0$$

$\Phi(x,y)$

$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y} - Xy - Yx$$

condiciones

función de tensión de Diry

Fuerzas de cuerpo / Fuerzas por unidad de volumen
 $F = \begin{bmatrix} X \\ Y \end{bmatrix} \rightarrow$ son constantes en (x,y)

$$\begin{bmatrix} \tau_{xy} & \tau_{yx} \\ \tau_{xy} & \tau_{yx} \end{bmatrix}$$

$$1) \frac{\partial^3 \Phi}{\partial x \partial y^2} + \frac{\partial}{\partial y} \left(-\frac{\partial^2 \Phi}{\partial x \partial y} - Xy - Yx \right) + X = 0$$

$$\frac{\partial^3 \Phi}{\partial x \partial y^2} - \frac{\partial^3 \Phi}{\partial x \partial y^2} - X + X = 0$$

$\Rightarrow$ En equilibrio

$$2) -\frac{\partial^3 \Phi}{\partial x^2 \partial y} - Y + \frac{\partial^3 \Phi}{\partial y \partial x^2} + Y = 0$$

a. $\Phi(x,y) = ax^2 + bxy + cy^2$

i. $a \neq 0, b = c = 0 \rightarrow$ estados tensional de punto

ii. $b \neq 0, a = c = 0 \rightarrow$

iii. $c \neq 0, a = b = 0 \rightarrow$

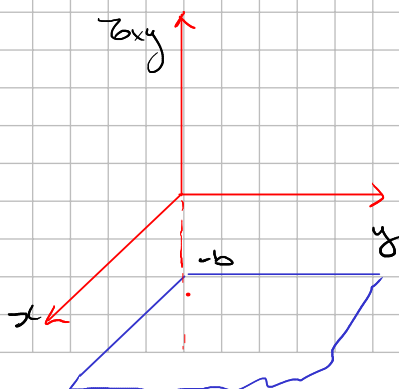
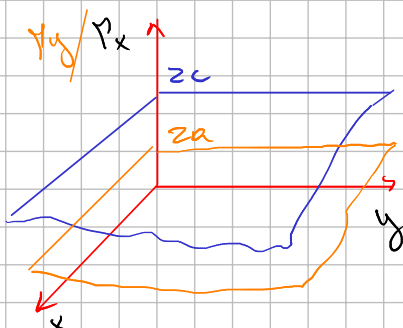
$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y} - Xy - Yx$$

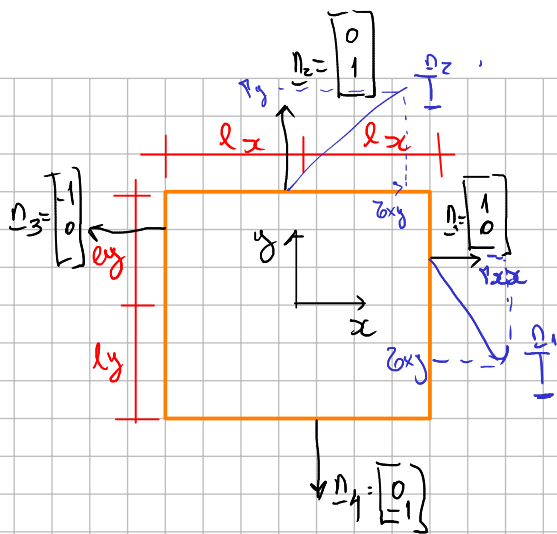
$$\frac{\partial \Phi}{\partial x} = 2ax + by$$

$$\frac{\partial \Phi}{\partial y} = bx + 2cy$$

$$\begin{cases} \tau_{xy} = 2c = \frac{\partial}{\partial y} (bx + 2cy) = 2c \\ \tau_{yx} = 2a \\ \tau_{xy} = -b - Xy - Yx \end{cases}$$

$$\begin{matrix} x=0 \\ y=0 \end{matrix} \Rightarrow \tau(x,y) = \begin{bmatrix} 2c & -b \\ -b & 2a \end{bmatrix}$$





$$\underline{T}(x, y) = \underline{\sigma}(x, y) \cdot \underline{n} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \end{bmatrix}$$

$$\underline{T}_1 = \begin{bmatrix} \sigma_{xx}(x, y) & \tau_{xy}(x, y) \\ \tau_{xy}(x, y) & \sigma_{yy}(x, y) \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \sigma_{xx}(l_x, y) \\ \tau_{xy}(l_x, y) \end{bmatrix}$$

$$\underline{T}_2 = \begin{bmatrix} \tau_{xy}(x, l_y) \\ \sigma_{yy}(x, l_y) \end{bmatrix}$$

$$\underline{T}_3 = - \begin{bmatrix} \sigma_{xx}(-l_x, y) \\ \tau_{xy}(-l_x, y) \end{bmatrix}$$

$$\underline{T}_4 = - \begin{bmatrix} \tau_{xy}(x, -l_y) \\ \sigma_{yy}(x, -l_y) \end{bmatrix}$$

a) ~~caso~~ i $a \neq 0, b = 0, c = 0$

$$\underline{F} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left. \begin{array}{l} \sigma_{xx} = 2c = 0 \\ \sigma_{yy} = 2a = 2a \\ \tau_{xy} = -b = 0 \end{array} \right\} \Rightarrow \underline{\sigma}(x, y) = \begin{bmatrix} 0 & 0 \\ 0 & 2a \end{bmatrix}$$

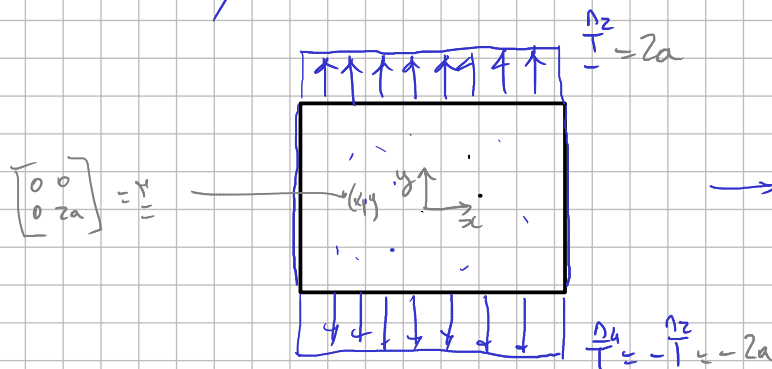
$$\underline{T}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

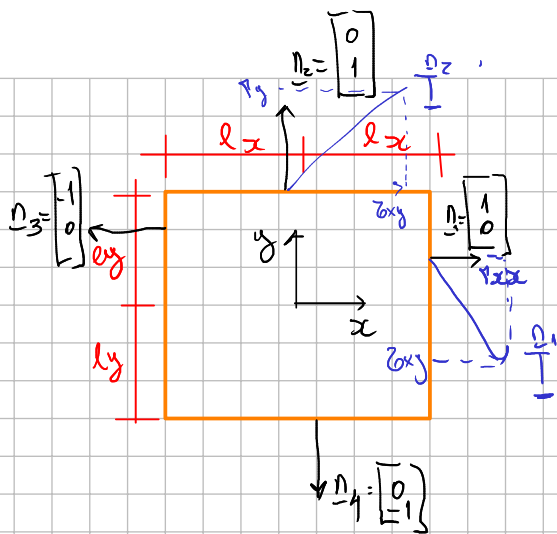
$$\underline{T}_2 = \begin{bmatrix} 0 \\ 2a \end{bmatrix}$$

$$\underline{T}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\underline{T}_4 = \begin{bmatrix} 0 \\ -2a \end{bmatrix}$$

$a > 0 \Rightarrow$ tracción z





$$\underline{T}(x, y) = \underline{\sigma}(x, y) \cdot \underline{n} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \end{bmatrix}$$

$$\underline{T}_1 = \begin{bmatrix} \sigma_{xx}(x, y) & \tau_{xy}(x, y) \\ \tau_{xy}(x, y) & \sigma_{yy}(x, y) \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \sigma_{xx}(l_x, y) \\ \tau_{xy}(l_x, y) \end{bmatrix}$$

$$\underline{T}_2 = \begin{bmatrix} \tau_{xy}(x, l_y) \\ \sigma_{yy}(x, l_y) \end{bmatrix}$$

$$\underline{T}_3 = - \begin{bmatrix} \sigma_{xx}(-l_x, y) \\ \tau_{xy}(-l_x, y) \end{bmatrix}$$

$$\underline{T}_4 = - \begin{bmatrix} \tau_{xy}(x, -l_y) \\ \sigma_{yy}(x, -l_y) \end{bmatrix}$$

a) ~~caso i~~ $a \neq 0, b = 0, c = 0$

$$\underline{F} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left. \begin{array}{l} \sigma_{xx} = 2c = 0 \\ \sigma_{yy} = 2a = 2a \\ \tau_{xy} = -b = 0 \end{array} \right\} \Rightarrow \underline{\sigma}(x, y) = \begin{bmatrix} 0 & 0 \\ 0 & 2a \end{bmatrix}$$

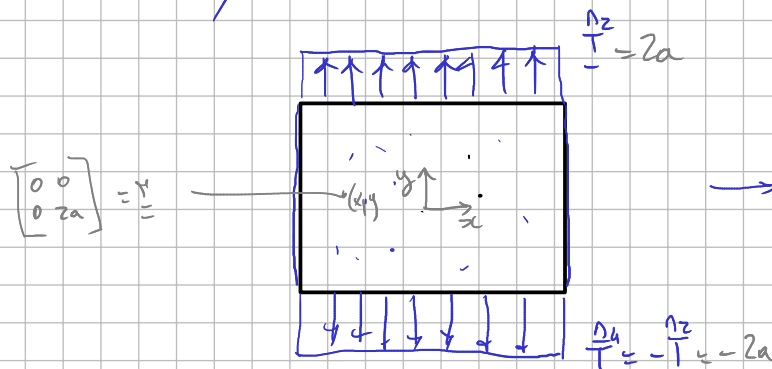
$$\underline{T}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\underline{T}_2 = \begin{bmatrix} 0 \\ 2a \end{bmatrix}$$

$$\underline{T}_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

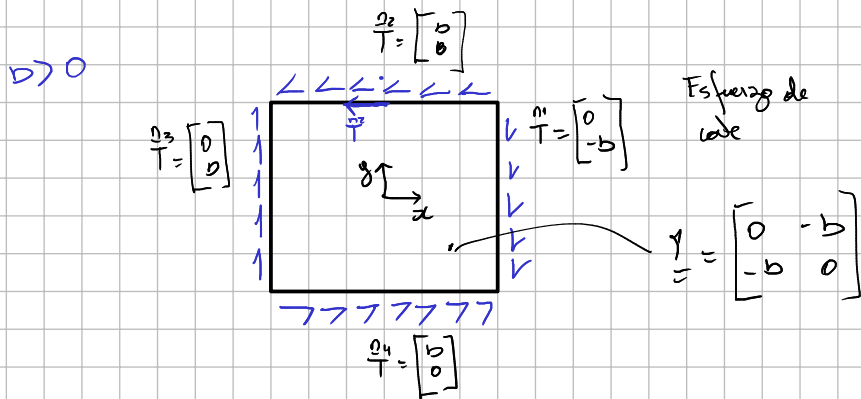
$$\underline{T}_4 = \begin{bmatrix} 0 \\ -2a \end{bmatrix}$$

$a > 0 \Rightarrow$ tracción z



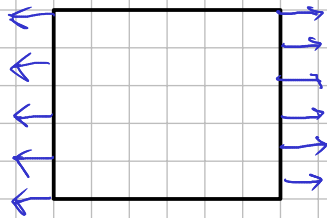
cas ii) $b \neq 0, a = c = 0 \Rightarrow \underline{\gamma} = \begin{bmatrix} 0 & -b \\ -b & 0 \end{bmatrix}$

$\frac{n_1}{T} = \begin{bmatrix} 0 \\ -b \end{bmatrix}, \frac{n_3}{T} = \begin{bmatrix} 0 \\ b \end{bmatrix}, \frac{n_2}{T} = \begin{bmatrix} -b \\ 0 \end{bmatrix}, \frac{n_4}{T} = \begin{bmatrix} b \\ 0 \end{bmatrix}$



cas iii) $c \neq 0, a = 0, b = 0 \Rightarrow \underline{\tau} = \begin{bmatrix} 2c & 0 \\ 0 & 0 \end{bmatrix}$

$c > 0$



Esf. $\propto |x|$ en direx. x

b. $\Phi(x, y) = ax^3 + bx^2y + cxy^2 + dy^3$

i. $d \neq 0, a = b = c = 0$

ii. $a \neq 0, b = c = d = 0$

iii. $b \neq 0, a = c = d = 0$

$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2}, \quad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2}, \quad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y} - \underbrace{X_y}_{0} - \underbrace{Y_x}_{0}$$

$$\frac{\partial \Phi}{\partial x} = 3ax^2 + 2bxy + cy^2$$

$$\frac{\partial \Phi}{\partial y} = bx^2 + 2cxy + 3dy^2$$

$$\vec{T} = \begin{cases} T_x = 2cx + 6dy \\ T_y = 6ax + 2by \end{cases}$$

$$T_{xy} = -2bx - 2cy$$

$$\vec{T}(x, y) = \begin{bmatrix} 6dy & 0 \\ 0 & 0 \end{bmatrix} \quad T_{xx}(x, y) = 6dy$$

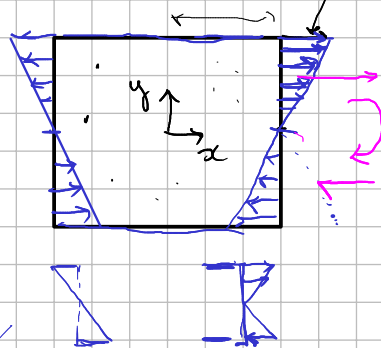
$$\frac{n_1}{I} = \begin{bmatrix} 6dy \\ 0 \end{bmatrix}$$

$$d > 0$$

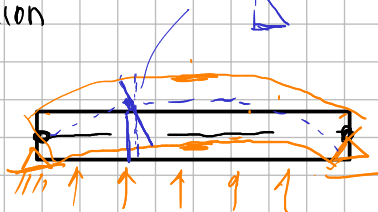
$$\frac{n_2}{I} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\frac{n_3}{I} = \begin{bmatrix} -6dy \\ 0 \end{bmatrix}$$

$$\frac{n_4}{I} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Flexión



caso iii) $b \neq 0, a = c = d = 0$

$$\vec{T} = \begin{cases} T_x = 2cx + 6dy \\ T_y = 6ax + 2by \end{cases}$$

$$\sigma_{xy} = -2bx - 2cy$$

$$T_x = 0$$

$$T_y = 2by$$

$$\sigma_{xy} = -2bx$$

$$\vec{T} = \begin{bmatrix} 0 & -2bx \\ 2bx & 2by \end{bmatrix}$$

$$b > 0 \quad \frac{n_2}{T} = \vec{T} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2bx \\ 2by \end{bmatrix} = \begin{bmatrix} -2bx \\ 2bly \end{bmatrix}$$

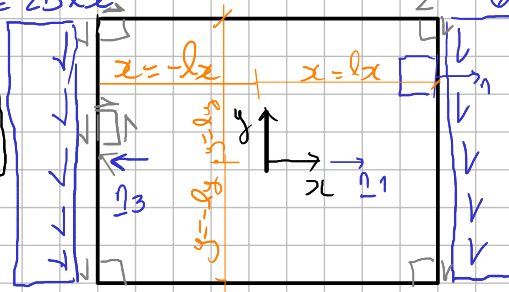
$$\vec{T} = (x, y) = \begin{bmatrix} 0 & 2bly \\ 2bly & 2by \end{bmatrix}$$

$$\sigma_{xy} = -2b(-lx) = 2bly$$

$$\sigma_{xy} = -2b \cdot lx$$

$$\frac{n_1}{T} = \vec{T} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -2bx \end{bmatrix} = \begin{bmatrix} 0 \\ -2bly \end{bmatrix}$$

$$\frac{n_3}{T} = \vec{T} \cdot \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2bx \end{bmatrix} = \begin{bmatrix} 0 \\ -2bly \end{bmatrix}$$



$$T_y = 2b(-ly) = -2bly$$

$$\sigma_{xy} = -2bx$$

$$\frac{n_4}{T} = \vec{T} \cdot \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2bx \\ -2by \end{bmatrix} = \begin{bmatrix} 2bx \\ -2b(-ly) \end{bmatrix} = \begin{bmatrix} 2bx \\ 2bly \end{bmatrix}$$

La estructura tiene que verificar el equilibrio global:

$$\sum F_x = 0 \Rightarrow \int_T T_x ds = 0 \Rightarrow$$

frontera

$$2b \cdot lx \cdot \frac{lx}{2} - 2bly \cdot \frac{lx}{2} - 2bly \cdot \frac{lx}{2} + 2bly \cdot \frac{lx}{2} = 0 \Rightarrow B.C.$$

$$\sum F_y = 0 \Rightarrow \int_r T_y ds = 0$$



$$-2bly \cdot 2lx - 2bly \cdot 2lx + 2bly \cdot 2lx + 2bly \cdot 2lx = 0 \Rightarrow B.C.$$

$$\sum M_z = 0 \Rightarrow \int_r (\underline{r} \times \underline{T}) \cdot \underline{k} ds$$

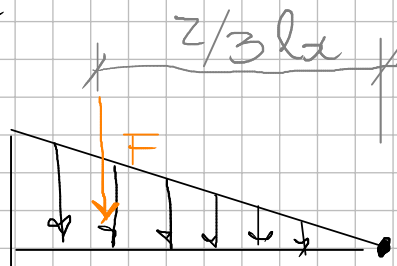
$$\begin{vmatrix} i & j & k \\ x & y & 0 \\ T_x & T_y & 0 \end{vmatrix} = (xT_y - yT_x)\underline{k}$$

$$= \int_r (x \cdot \frac{dT_y}{ds} ds - y \cdot \frac{dT_x}{ds} ds) =$$

$$= \underbrace{2bly \cdot 2lx \cdot 0}_{\frac{F}{2} \cdot d} + \underbrace{2bly \cdot 2lx \cdot 0}_{\frac{F}{2} \cdot d} + \underbrace{2bly \cdot 2lx \cdot lx}_{\frac{F}{2} \cdot d} - \underbrace{2bly \cdot 2lx \cdot lx}_{\frac{F}{2} \cdot d} -$$

$$- \underbrace{2bly \cdot lx \cdot ly}_{\frac{F}{2} \cdot d} + \underbrace{2bly \cdot lx \cdot ly}_{\frac{F}{2} \cdot d} - \underbrace{2bly \cdot lx \cdot ly}_{\frac{F}{2} \cdot d} + \underbrace{2bly \cdot lx \cdot ly}_{\frac{F}{2} \cdot d} =$$

$$= 0 \Rightarrow B.C.$$



$$M_z = F \cdot d = F \cdot \frac{2}{3} lx$$